

Heart Disease Risk Prediction with Explainable AI: Counterfactual-Based Actionable Health Recommendations

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1 Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, claiming approximately 17.9 million lives each year [1]. A significant percentage of these deaths are preventable through early identification of risk factors such as high cholesterol, smoking, and physical inactivity.

Machine learning models have demonstrated strong performance in predicting CVD risk from patient health metrics, but those prediction alone is not enough for actual clinical recommendations. Traditional explainable AI methods such as SHAP [2] and LIME [3] answer why did the model predict this by attributing feature importances, rather than explaining what actions to take. Counterfactual explanations [4] address this by finding the smallest change to your metrics that would flip the prediction from high-risk to low-risk.

In this work, we train and compare five ML models (Logistic Regression, SVM, KNN, Random Forest, XGBoost) with F_2 -score optimization. Then we design a domain-constrained counterfactual generator that respects domain knowledge such as immutable features, directional constraints, and human effort costs. Finally we deploy the pipeline as a Streamlit web application with real-time risk prediction and visual action plans.

2 Dataset and Feature Engineering

2.1 Dataset Overview

We use a synthetic Kaggle healthcare dataset of $N = 15,000$ patient records with a binary target variable `Heart_Disease_Risk` $\in \{0, 1\}$. This raw dataset contains 16 features organized into four categories (Table 1).

Table 1: Raw feature descriptions by clinical category.

Category	Feature	Type
Demographics	Age	Continuous
	Gender	Binary
	Height_cm	Continuous
	Weight_kg	Continuous
Vitals & Labs	Systolic_BP	Continuous (mmHg)
	Diastolic_BP	Continuous (mmHg)
	Cholesterol_Total	Continuous (mg/dL)
	Cholesterol_HDL	Continuous (mg/dL)
	Cholesterol_LDL	Continuous (mg/dL)
	Fasting_Blood_Sugar	Continuous (mg/dL)
Lifestyle	Smoking_Status	Binary
	Alcohol_Consumption	Ordinal (0-3)
	Physical_Activity_Level	Ordinal (0-4)
	Stress_Level	Ordinal (0-10)
	Sleep_Hours	Discrete (3-12)
Genetics	Family_History	Binary

2.2 Exploratory Data Analysis

Prior to modeling, we performed EDA to understand feature distributions and their relationship with the target. The target variable exhibits moderate class imbalance, motivating class-weighted training. Continuous features (Systolic_BP, Cholesterol_Total, Fasting_Blood_Sugar) follow approximately normal distributions, while lifestyle features are discrete and bounded. Pearson correlation analysis reveals moderate inter-correlation among blood pressure readings ($r \approx 0.3-0.5$) and cholesterol subtypes, which informed the design of the derived features. Patients classified as high-risk tend to have elevated systolic blood pressure, higher LDL cholesterol, lower physical activity, and a family history of heart disease.

2.3 Derived Features

We then engineer 5 additional features from the raw inputs using actual medical formulas:

$$\text{BMI} = \frac{\text{Weight_kg}}{(\text{Height_cm}/100)^2} \quad (1)$$

$$\text{Cholesterol_Ratio} = \frac{\text{Cholesterol_Total}}{\text{Cholesterol_HDL}} \quad (2)$$

$$\text{LDL_HDL_Ratio} = \frac{\text{Cholesterol_LDL}}{\text{Cholesterol_HDL}} \quad (3)$$

$$\text{Pulse_Pressure} = \text{Systolic_BP} - \text{Diastolic_BP} \quad (4)$$

$$\text{MAP} = \text{Diastolic_BP} + \frac{\text{Pulse_Pressure}}{3} \quad (5)$$

2.4 Preprocessing

We standardized the features using z-score normalization: $x_i^{\text{scaled}} = (x_i - \mu_i)/\sigma_i$, where μ_i and σ_i are computed on the training set only to avoid data leakage. The scaler is reused at inference time.

3 Methodology

3.1 Training Pipeline

All five models use the following pipeline: First they will go through splitting by stratified 80/20 train-test split, then the StandardScaler preprocessing, and GridSearchCV with StratifiedKFold ($k=5$) for hyperparameter optimization.

In medical risk prediction, missing a high-risk patient (false negative) is far more dangerous than flagging a healthy one (false positive). We use the F_2 -score as the optimization metric for this reason. The F_β -score weights recall β^2 times more than precision:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} \quad (6)$$

With $\beta=2$, recall is weighted 4 times higher than precision, prioritizing detection of at-risk patients. To address class imbalance, we use balanced class weights ($w_c = N/2N_c$) for LR, SVM, and RF, and equivalent scale_pos_weight for XGBoost.

3.2 Model Architectures

3.2.1 Logistic Regression

Logistic Regression [5] models the positive class probability using the Sigmoid function:

$$P(y=1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (7)$$

where $\mathbf{w} \in \mathbb{R}^d$ is the coefficient vector and b is the intercept. Training minimizes the L_2 -regularized cross-entropy loss with inverse regularization strength $C \in \{0.01, 0.1, 1.0, 10.0\}$. A key advantage for xAI is that each w_i directly represents the change in log-odds per unit increase in feature x_i , enabling coefficient extraction for fast counterfactual inference (Section 3.3).

3.2.2 Support Vector Machine

SVM [6] with an RBF kernel finds the optimal separating hyperplane in kernel space:

$$f(\mathbf{x}) = \sum_{i=1}^{N_{SV}} \alpha_i y_i \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}\|^2) + b \quad (8)$$

where α_i are learned support vector weights and γ controls kernel bandwidth. We search $C \in \{0.1, 1.0, 10.0\}$ and $\gamma \in \{\text{scale}, \text{auto}\}$, with Platt scaling for probability calibration.

3.2.3 K-Nearest Neighbors

KNN [7] classifies by majority vote among the k nearest neighbors using the Minkowski distance:

$$d_p(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{l=1}^d |x_{il} - x_{jl}|^p \right)^{1/p} \quad (9)$$

We search $k \in \{3, 5, 7, 9, 11\}$, weights $\in \{\text{uniform}, \text{distance}\}$, and $p \in \{1, 2\}$.

3.2.4 Random Forest

Random Forest [8] aggregates T decision trees, each trained on a bootstrap sample with random feature subsets. The final prediction is the majority vote $\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T$, where each tree splits on the feature minimizing Gini impurity $G = 1 - \sum_c p_c^2$. We search $T \in \{50, 100, 200\}$, max depth $\in \{5, 10, \text{None}\}$, and min samples per leaf $\in \{1, 2, 4\}$.

3.2.5 XGBoost

XGBoost [9] builds gradient-boosted trees sequentially, where each new tree f_t corrects the residuals of the previous ensemble: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i)$. The objective balances loss with a complexity regularizer:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \gamma T_{\text{leaves}} + \frac{\lambda}{2} \sum_j w_j^2 \quad (10)$$

We search $T \in \{50, 100, 200\}$, max depth $\in \{3, 5, 7\}$, and learning rate $\eta \in \{0.01, 0.1, 0.2\}$.

3.3 Counterfactual Explanation System

3.3.1 Problem Formulation

Given a classifier f and a patient $\mathbf{x}_0$ with $f(\mathbf{x}_0) = 1$ (high-risk), we seek a counterfactual $\mathbf{x}'$ such that: (i) $f(\mathbf{x}') = 0$, (ii) the cost-weighted distance is minimized, (iii) derived features are physically consistent, and (iv) domain constraints are satisfied.

3.3.2 Domain Constraints

We encode five constraint categories. **Immutable features** $\mathcal{I} = \{\text{Age, Gender, Height, Family_History}\}$ cannot be changed. **Derived features** $\mathcal{D}$ (BMI, cholesterol ratios, pulse pressure, MAP) are auto-recomputed from source features using Equations (1)-(5), never directly optimized. The 12 remaining **source features** $\mathcal{S} = \mathcal{F} \setminus (\mathcal{I} \cup \mathcal{D})$ are the optimization variables.

Each source feature has a **directional constraint**: blood pressure, cholesterol, blood sugar, smoking, alcohol, and stress can only decrease; physical activity, sleep, and HDL can only increase. Each is also assigned a **cost weight** c_i reflecting human effort: low for behavioral changes (e.g., $c_{\text{alcohol}} = 1.0$), high for metabolic changes requiring medication (e.g., $c_{\text{cholesterol}} = 5.0$).

3.3.3 Objective Function

The optimizer minimizes a delta vector $\boldsymbol{\delta} \in \mathbb{R}^{12}$:

$$\min_{\boldsymbol{\delta}} \underbrace{\sum_{i \in \mathcal{S}} c_i \cdot \frac{|\delta_i|}{r_i}}_{\text{Human Effort Cost}} + \underbrace{100 \cdot \max(0, P(y=1 | \mathbf{x}_0 + \boldsymbol{\delta}) - 0.30)}_{\text{Prediction Penalty}} \quad (11)$$

subject to box constraints $\ell_i \leq \delta_i \leq u_i$ encoding directional rules and data-range limits. The threshold $\tau=0.30$ provides a safety margin below the 0.50 decision boundary, ensuring robustness to post-processing rounding.

3.3.4 Optimization and Post-Processing

We use **L-BFGS-B** [10], a quasi-Newton optimizer that natively enforces box constraints via gradient projection. To escape local minima, we use a multi-start strategy ($R=4$): one run from $\boldsymbol{\delta}_0 = \mathbf{0}$ and three from random points within 20% of bounds, keeping the best result.

After convergence, lifestyle features are rounded to integers, continuous features to 1 decimal, and derived features are recomputed. Changes below clinically meaningful thresholds (e.g., $\Delta \text{BP} < 5$ mmHg) are discarded. The rounded solution is re-evaluated to confirm prediction flip.

For real-time performance, we compute predictions directly via pre-extracted coefficients:

$$P(y=1 | \mathbf{x}) = \sigma\left(\mathbf{w}^\top \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}} + b\right) \quad (12)$$

4 Experimental Results

4.1 Model Comparison

Table 2: Model performance on the 20% holdout test set.

Model	Accuracy	Precision	Recall	F ₂
Logistic Regression	0.7283	0.6814	0.7080	0.7025
SVM (RBF)	0.7193	0.6713	0.6980	0.6925
KNN	0.7130	0.6959	0.6070	0.6229
Random Forest	0.7153	0.6667	0.6942	0.6885
XGBoost	0.7243	0.6793	0.6965	0.6930

Note: Fill in from evaluation notebook output.

4.2 Model Selection for Deployment

We select **Logistic Regression** for deployment despite potentially higher accuracy from ensemble models, for three reasons: (1) coefficient transparency enables direct interpretation of each feature’s risk contribution, (2) the closed-form sigmoid allows 50× faster counterfactual inference critical for real-time deployment, and (3) linear log-odds ensure monotonic feature effects, preventing paradoxical suggestions where reducing blood pressure increases predicted risk (a known issue with tree-based models due to axis-aligned splits).

4.3 Counterfactual Quality

Table 3: Example counterfactual action plan for a high-risk patient.

Feature	Original	Target	Δ	Type
Smoking Status	Smoker	Non-Smoker	−1	Lifestyle
Physical Activity	1	3	+2	Lifestyle
Stress Level	7	4	−3	Lifestyle
Sleep Hours	5	7	+2	Lifestyle
Systolic BP	160	145	−15	Medical

The optimizer prioritizes low-cost lifestyle changes before recommending medical interventions. We evaluated the DiCE framework [11] as an alternative, but found it produces physically inconsistent patients (e.g., low weight with high BMI), lacks directional constraints, and does not model human effort costs.

5 System Architecture

The deployed Streamlit application has three layers: (1) a data layer with persisted scaler and feature names, (2) a model layer with five serialized classifiers (LR used at inference), and (3) an application layer with patient input forms, Plotly risk gauge charts (before and after), a horizontal bar chart comparing current versus target values, and metric cards with delta arrows for each recommended action. All assets are cached for instant page loads.

6 Discussion and Conclusion

This system shifts from *explanatory* to *actionable* AI. Unlike SHAP/LIME which explain *why* a prediction was made, our counterfactual system tells patients *what to do*. Domain constraints ensure medical realism, cost weights prioritize easy changes first, and coefficient extraction enables sub-second response times.

Limitations include the use of synthetic training data, LR’s assumption of linear decision boundaries, manually assigned cost weights, and L-BFGS-B’s local optimality. Future work includes validation on real clinical datasets (e.g., Framingham Heart Study), empirically learning cost weights from patient adherence data, extending the explainer to non-linear models, and integrating SHAP alongside counterfactuals to provide both “why” and “what to do” in one interface.

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